

DSAA3073: Theories in Data Science

Week 8 Open Investigation

When May an Ensemble Certificate Be Selected?

A clinical audit, a fixed-threshold theorem, and two altered assumptions

Basic analysis. A sepsis early-warning audit

Use only the margin and AdaBoost results established in the Week 8 Learning Sheet

A hospital research group has fitted an ensemble of binary bedside rules to flag patients who may require urgent sepsis review. Before deployment, the group evaluates the fitted score on an i.i.d. audit sample of $m = 120000$ labelled encounters. The base-rule class $\mathcal{H}$ is binary: every $h \in \mathcal{H}$ satisfies $h : \mathcal{X} \rightarrow \{-1, +1\}$. Its VC dimension is $d = 10$. Write the fitted score as $F(x) = \sum_j \alpha_j h_{j(x)}$, where $\alpha_j \geq 0$ and $\|\alpha\|_1 = \sum_j \alpha_j > 0$. Define its normalized ensemble explicitly by

$$\bar{F}(x) = \frac{F(x)}{\|\alpha\|_1} = \sum_j \frac{\alpha_j}{\|\alpha\|_1} h_{j(x)} \in \text{conv}(\mathcal{H}).$$

The normalized margin is $\rho(x, y) = y\bar{F}(x)$. The empirical low-margin fractions are

$$\hat{R}_{S,0.20} = 0.065, \quad \hat{R}_{S,0.40} = 0.150.$$

The audit committee requires confidence $1 - \delta = 0.95$. A smaller certified upper bound is preferred, but the certificate is not a substitute for the hospital's prospective safety study.

Basic Task 1. Construct and compare the two population certificates

State the normalized functional margin and the fixed-threshold Week 8 theorem. Before using it, verify each condition explicitly: the audit sample is i.i.d.; the base class is binary; $1 \leq d \leq m$; the coefficient normalization satisfies $\|\alpha\|_1 > 0$; the selected threshold is fixed and positive; and $0 < \delta < 1$. Instantiate the theorem separately at $\rho_0 = 0.20$ and $\rho_0 = 0.40$, calculate both numerical upper bounds, and recommend which certificate the committee should report. Explain why the larger threshold need not give the smaller certificate.

Basic analysis. One declared AdaBoost coordinate step

Separate the empirical optimization calculation from the population certificate

On the current normalized example distribution, a candidate bedside rule has weighted error $\varepsilon = 0.30$. The weak learner selects it as the minimum-error coordinate in the fixed binary dictionary.

Basic Task 2. Derive the step and state exactly what it establishes

Derive the exact positive line-search coefficient and the multiplicative change in empirical exponential loss. Then distinguish the optimization conclusion from the population statement in Basic Task 1 and from observed prospective clinical performance.

Thinking investigation. Formalize the certificate

Make the order of selection and probability explicit

Represent the analysis by

$$(\mathcal{A}, \mathcal{F}, \rho_0, \mathcal{G}),$$

where $\mathcal{A}$ describes the audit sampling, $\mathcal{F} = \text{conv}(\mathcal{H})$ contains the explicitly defined score $\bar{F} = \frac{F}{\|\alpha\|_1}$, ρ_0 is the margin threshold, and $\mathcal{G}$ is the high-probability population-error guarantee. In the basic certificate, $\mathcal{A}$ is i.i.d. sampling and ρ_0 is fixed before the confidence event; the event is uniform over $\bar{F} \in \mathcal{F}$, not over arbitrary data-dependent thresholds.

Investigation 1. Select the threshold after seeing the audit

Suppose the committee pre-registers a finite grid $\Gamma = \{\rho_1, \dots, \rho_K\}$ but chooses the most favorable threshold only after inspecting the empirical margin distribution. Does the original fixed-threshold confidence statement survive unchanged? Derive a valid simultaneous analytic certificate over the grid using only the theorem and a union bound. Give the resulting expression for $K = 20$ and explain the statistical price of selection.

Thinking investigation. Remove the exact line-search assumption

The AdaBoost identity fails, but the empirical penalty remains identifiable

Keep the selected coordinate and its weighted error $0 < \varepsilon < \frac{1}{2}$, but let an implementation take the inexact step $\eta = \alpha + s$, where α is the exact AdaBoost coefficient and s is a deterministic line-search error.

Investigation 2. Derive the exact cost of an inexact step

Does the update still coincide with the Week 8 AdaBoost step when $s \neq 0$? Starting from the correct/error mass decomposition, derive a closed-form expression for $Z(\alpha + s)$ and for its ratio to the optimal line value. Determine whether every finite inexact step still decreases the objective.

Synthesis. State the assumption-guarantee map

Explain, in one coherent paragraph, which conclusion is supported under each of the following regimes: one fixed threshold; a data-selected point on a pre-registered finite grid; exact AdaBoost coordinate and line rules; and an inexact line step. Identify which expressions are population certificates and which are empirical optimization identities.